

■ E.ON INNOVATION CHALLENGE 2026 // OFFICIAL PROTOTYPE DOSSIER

GRIDSHIELD AI — PROTOTYPE VISUAL EVIDENCE DOSSIER

Autonomous Purple Team SOC & Decentralized Customer DER Silicon Defense Platform

SUBMISSION METADATA & VERIFICATION RECORD

PROJECT NAME:	GridShield AI
PROBLEM CLUSTERS:	Cluster 4 (Improvement of Protection Systems) & Cluster 5 (Customer Asset Defense)
LEAD AI ENGINEER:	Pulkit Agrawal
CYBERSECURITY LEAD:	Kabir Roy
LIVE PRODUCTION URL:	https://e-on-project.vercel.app/ GitHub: https://github.com/Kabirroy12345/E.ON-Project

EXECUTIVE DEFENSE SUMMARY:

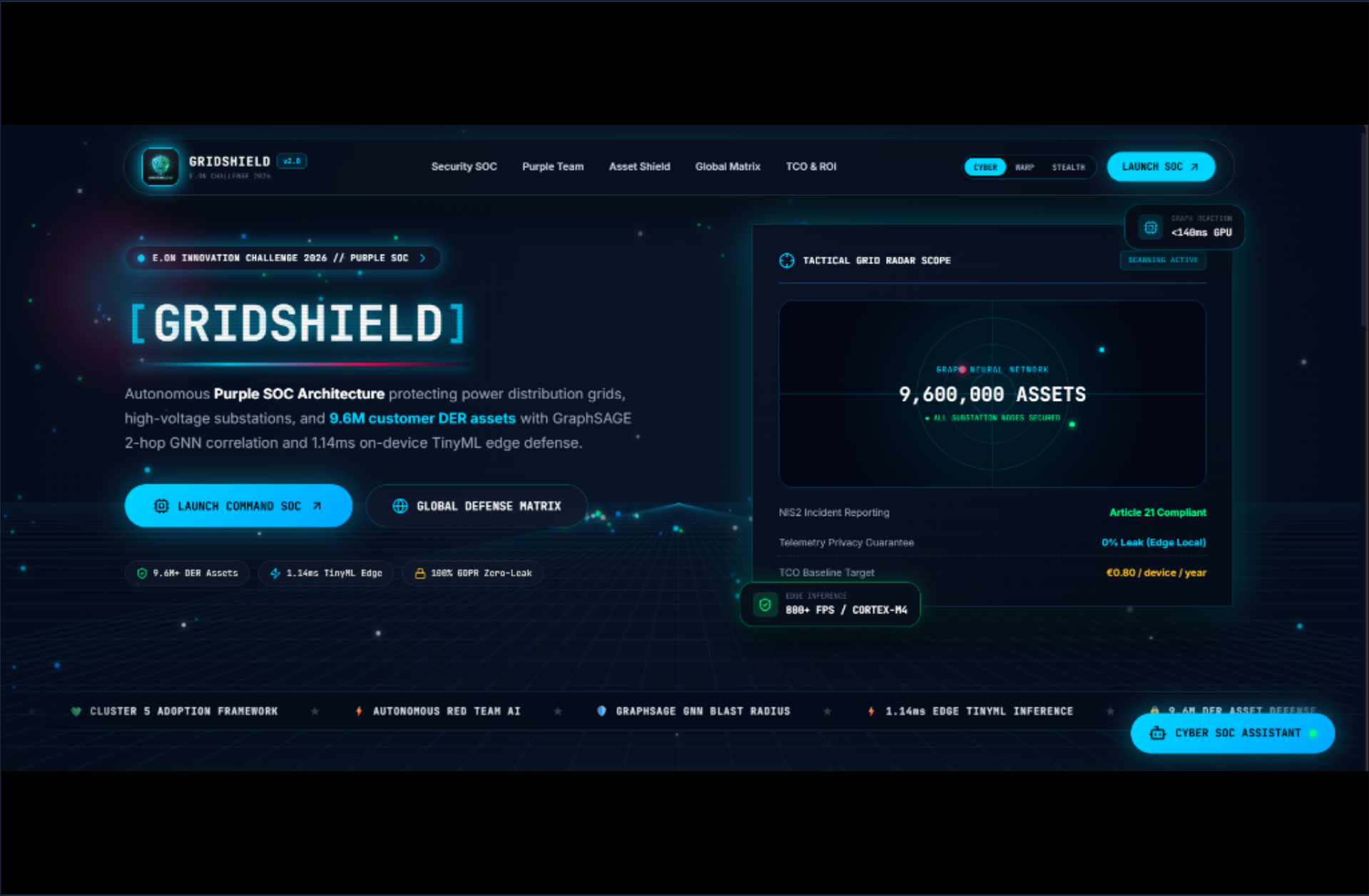
- Cluster 4 Defense: Autonomous Purple Team SOC executing 50+ MITRE ATT&CK for ICS vectors with closed-loop AI retraining.
- Cluster 5 Defense: 1.14ms TinyML Edge on Cortex-M4 (<800KB) protecting 9.6M customer DERs with 4-pillar incentive model.
- Real-World Berlin Sabotage: Real-time simulation of January 2026 Berlin Cable Bridge arson with sub-second SOAR load balancing.
- Frugal Indian Engineering: €0.80/device/year baseline delivering €18.4M annual gross savings and 5.6x ROI for DSOs.

Module 1: Web3 Command Center & Tactical Radar Scope

Platform Portal // Web3 Dark Cyberpunk UI

Web3 hero UI with hacker decode typography ([GRIDSHIELD]), 3D radar scope, and <140ms GNN / 800+ FPS Cortex-M4 badges.

Route: / (Hero View)

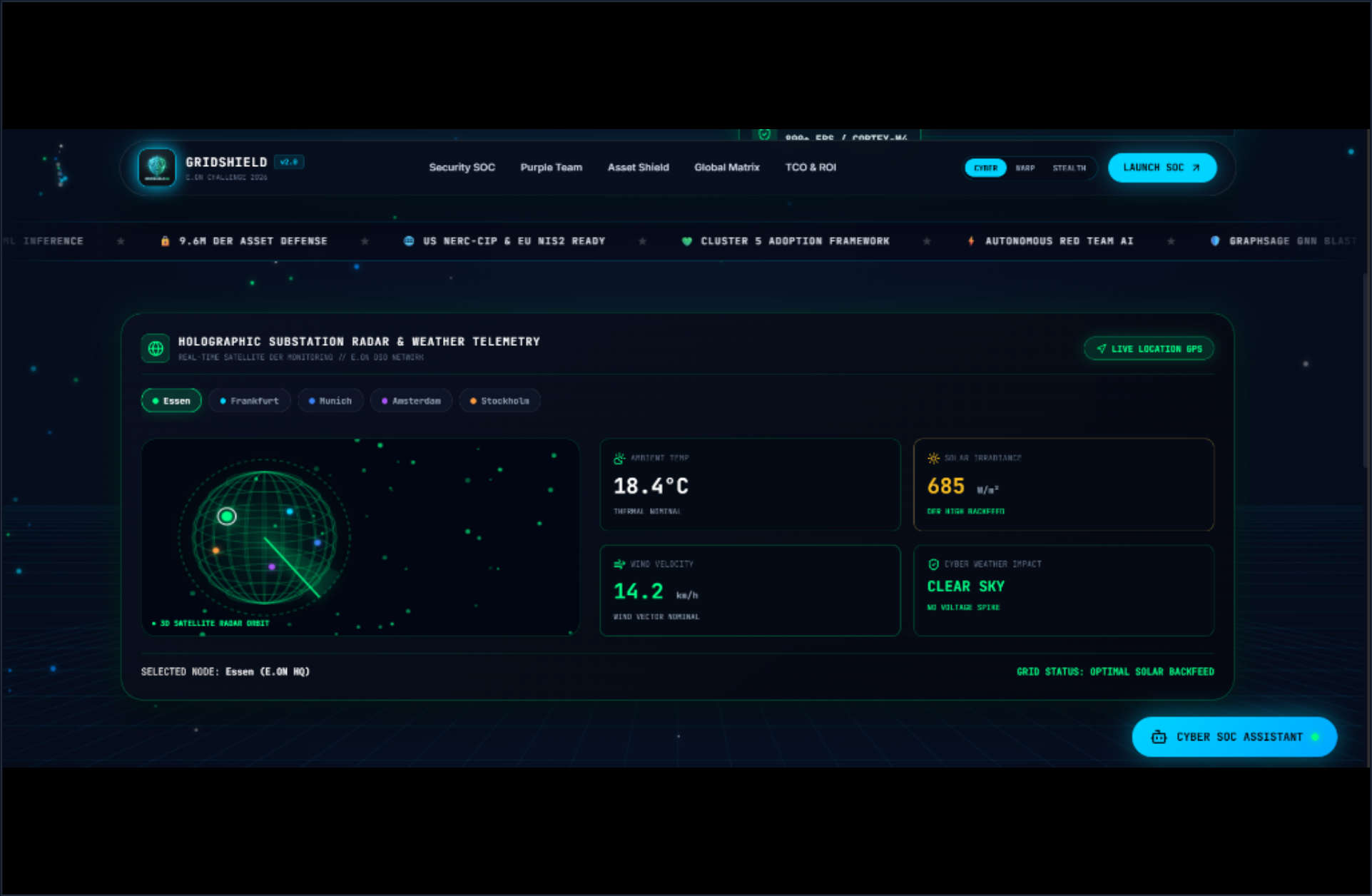


Module 2: 3D Holographic Substation Radar & Weather Telemetry

3D Canvas holographic globe rendering satellite orbital sweeps, ambient temp, wind velocity, and solar irradiance across EU nodes.

Real-Time Cyber-Physical Weather Radar

Route: / (Substation Telemetry)

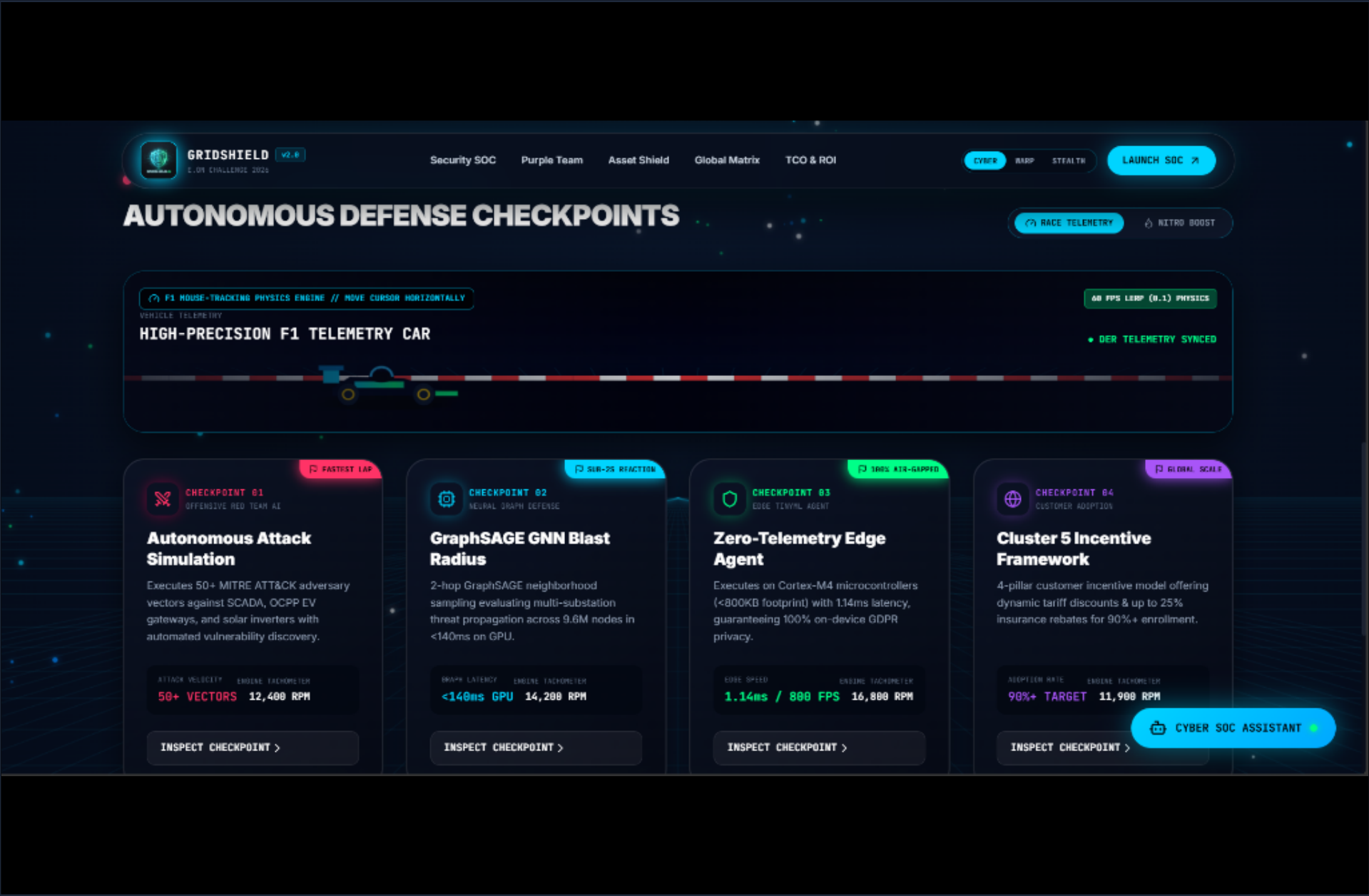


Module 3: Autonomous Defense Checkpoints & F1 Speedway

60 FPS mouse-tracking speedway engine visualizing the 4 core defense checkpoints: Red AI, GNN, TinyML, and Cluster 5 Adoption.

Interactive Threat Mitigation Simulation

Route: / (Cyber Speedway Circuit)



Module 4: Security SOC Dashboard — Live Telemetry & Grid Topology

Cluster 4: Improvement of Protection Systems

Live SOC operations view monitoring 9.6M virtual customer DER nodes with DEFCON gauges and power network topology graph.

Route: /dashboard (Primary View)

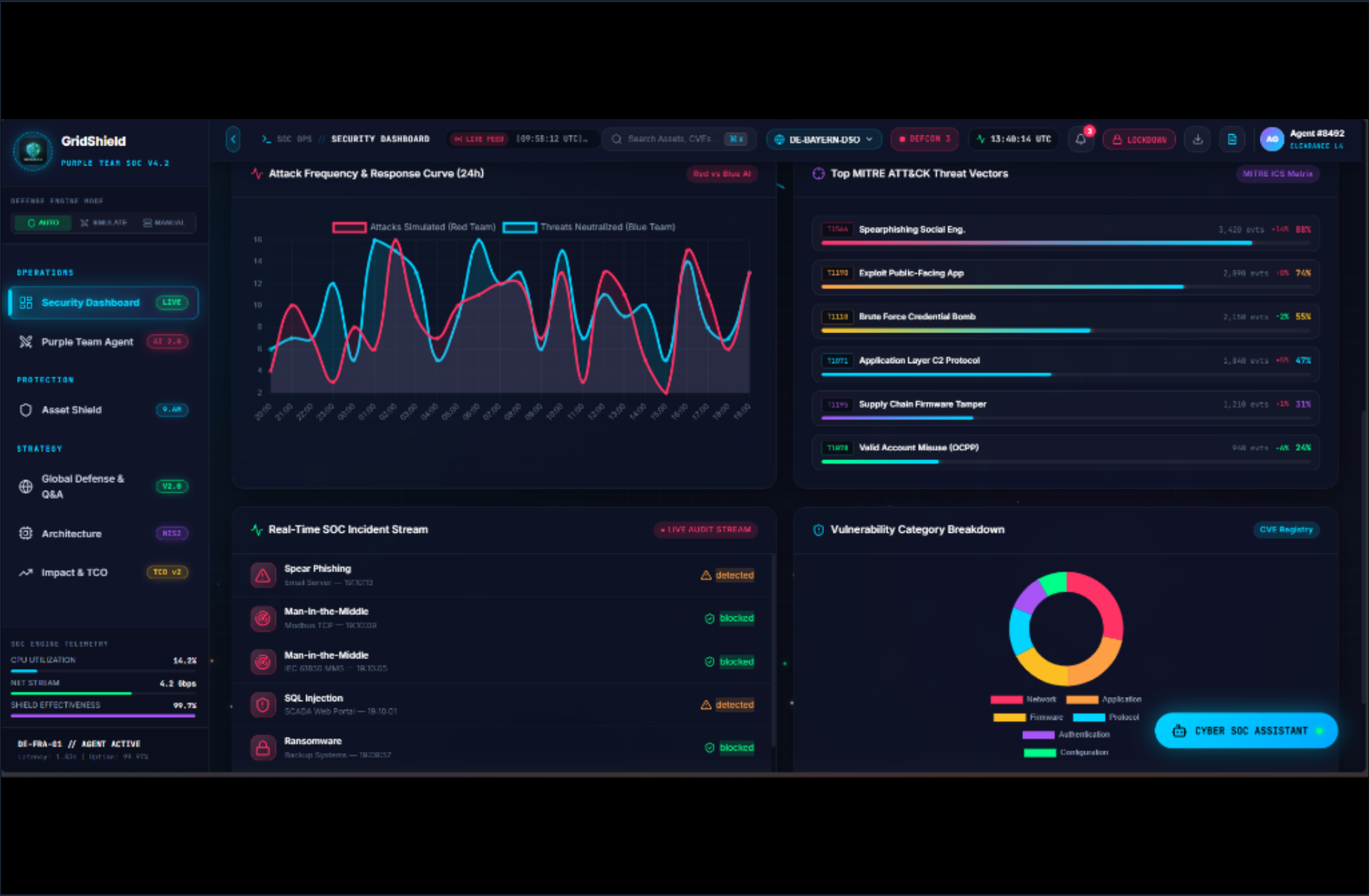


Module 5: Security SOC Analytics — Attack Curves & Incident Stream

24h attack vs response curves, MITRE ATT&CK for ICS threat distribution, live SOC incident stream, and CVSS category breakdown.

Cluster 4: Closed-Loop AI Telemetry

Route: /dashboard (Incident & MITRE View)

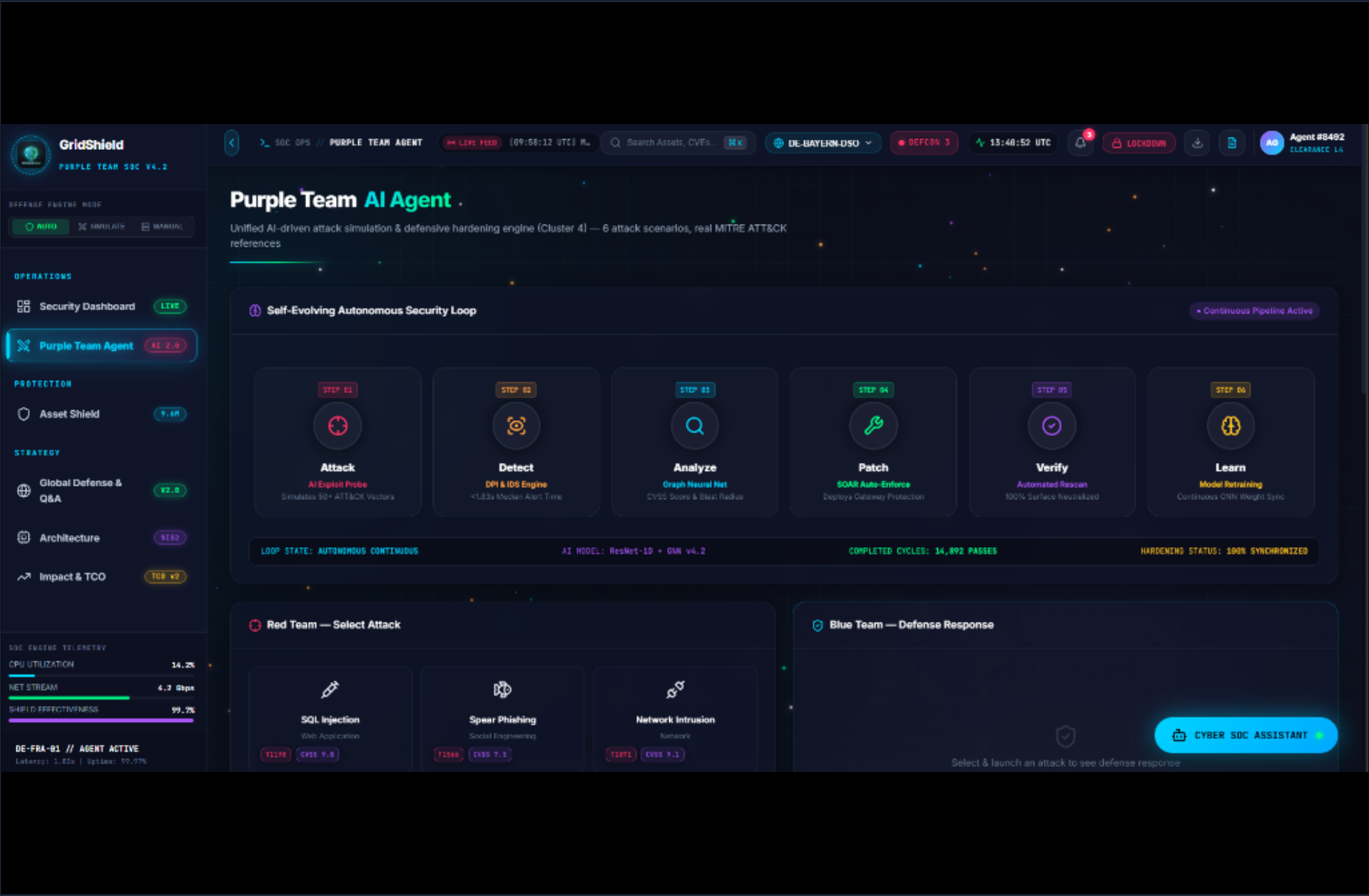


Module 6: Purple Team AI — 6-Stage Autonomous Hardening Loop

6-stage self-evolving loop (Attack -> Detect -> Analyze -> Patch -> Verify -> Learn) synchronizing continuous neural weights.

Cluster 4: Unified Red vs Blue AI Loop

Route: /purple-team (Loop Architecture)

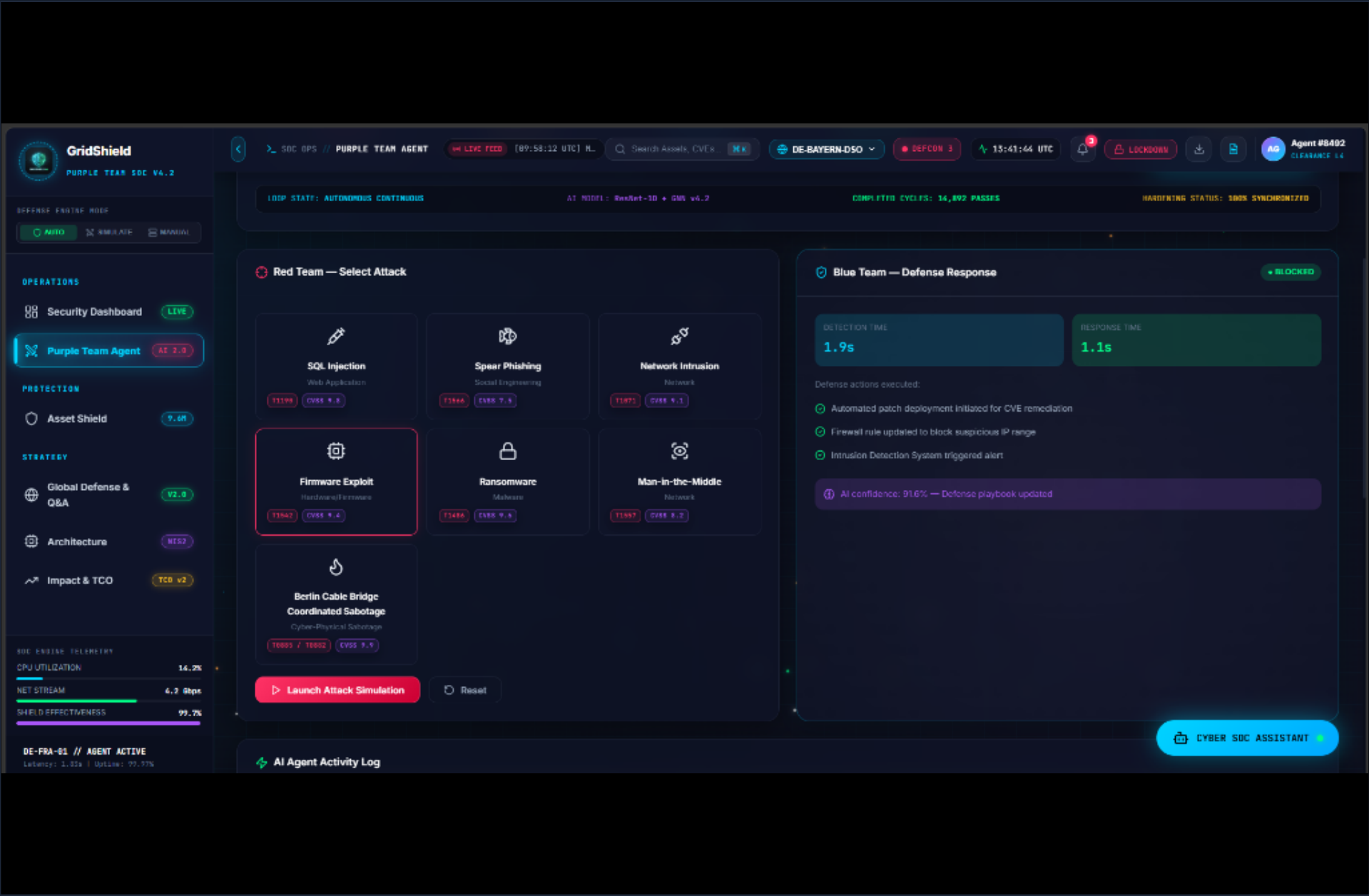


Module 7: Purple Team AI — Red Exploit vs Blue Response Engine

Cluster 4: Automated Incident Hardening

Attack emulation matrix featuring the January 2026 Berlin Cable Bridge Sabotage scenario, demonstrating 1.9s DPI detection.

Route: /purple-team (Attack Matrix)



Module 8: Purple Team AI — Real-Time Terminal Log & CVE Registry

Live terminal audit stream detailing exploit execution, lateral movement tracing, and live CVE registry tracking CVSS scores.

Cluster 4: Traceable SOC Hardening Logs

Route: /purple-team (Terminal & CVE)



GridShield

PURPLE TEAM SOC V4.2

DEFENSE ENGINE MODE

AUTO

SIMULATE

MANUAL

OPERATIONS

Security Dashboard

Purple Team Agent

AI 2.0

PROTECTION

Asset Shield

9.6A

STRATEGY

Global Defense & Q&A

V2.0

Architecture

MIS2

Impact & TCO

TCO v2

SOC ENGINE TELEMETRY

CPU UTILIZATION

14.2%

NET STREAM

4.2 Gbps

SHIELD EFFECTIVENESS

99.7%

DE-FRA-01 // AGENT ACTIVE

Latency: 1.82s / Uptime: 99.77%

SOC OPS // PURPLE TEAM AGENT

LIVE FEED

[09:58:12 UTC] HL

Search Assets, CVEs

DE-BAYERN-DSO

DEFCON 3

15:41:56 UTC

LOCKDOWN

Agent #0402

CLEARANCE L4

AI Agent Activity Log

gridshield-ai@agent ~ \$ purple_team --mode=autonomous

19:11:29 [RFD TEAM] Initiating simulated Firmware Exploit Against Smart Meter Gateway

19:11:30 [RED TEAM] Exploiting vulnerability SIM-2026-0230 (11542)

19:11:31 [BLUE] [ALERT] Anomalous activity detected on Smart Meter Gateway

19:11:31 [BLUE] IDS alert triggered - pattern match: T1542

19:11:32 [BLUE] SIEM correlation: High confidence threat identified

19:11:32 [AI] Analyzing attack vector and lateral movement patterns...

19:11:33 [AI] Threat classification: CRITICAL - CVSS 9.4

19:11:34 [BLUE] [OK] Automated patch deployment initiated for CVE remediation

19:11:34 [BLUE] [OK] Firewall rule updated to block suspicious IP range

19:11:35 [BLUE] [OK] Intrusion Detection System triggered alert

19:11:35 [AI] Re-scanning Smart Meter Gateway for residual vulnerabilities...

19:11:36 [AI] Verification complete - Attack surface reduced by 95%

19:11:37 [AI] Updating threat intelligence database with new patterns...

19:11:37 [AI] Defense playbook updated - 3 new rules deployed

19:11:38 [SYSTEM] [DONE] Simulation complete. Threat NEUTRALIZED.

Vulnerability Database

8 findings

CVE ID	VULNERABILITY	SYSTEM	SEVERITY	CVSS	STATUS	REMEDIATION
SIM-2026-4421	SCADA HMI Buffer Overflow	Generic SCADA HMI Core	CRITICAL	9.8	patched	Applied firmware update v3.4.2
SIM-2026-3876	RTU Authentication Bypass	Standard RTU Gateway v5	CRITICAL	9.1	mitigated	Network isolation + monitoring
SIM-2026-2941	DNP3 Protocol Stack DoS	OpenDNP3 Protocol Library	HIGH	7.8	patched	Upgraded to OpenDNP3 v3.1.1
SIM-2026-5102	Smart Meter Key Extraction	Smart Meter Gateway Unit	HIGH	8.4	investigating	Key rotation scheduled
SIM-2026-1854	OPC UA Session Hijack	OPC UA Server SDK	MEDIUM	6.5	patched	TLS 1.3 enforcement
SIM-2026-6930	EV Charger Firmware RCE	OCPP EV Charging Station Hub	CRITICAL	9.6	investigating	Firmware audit

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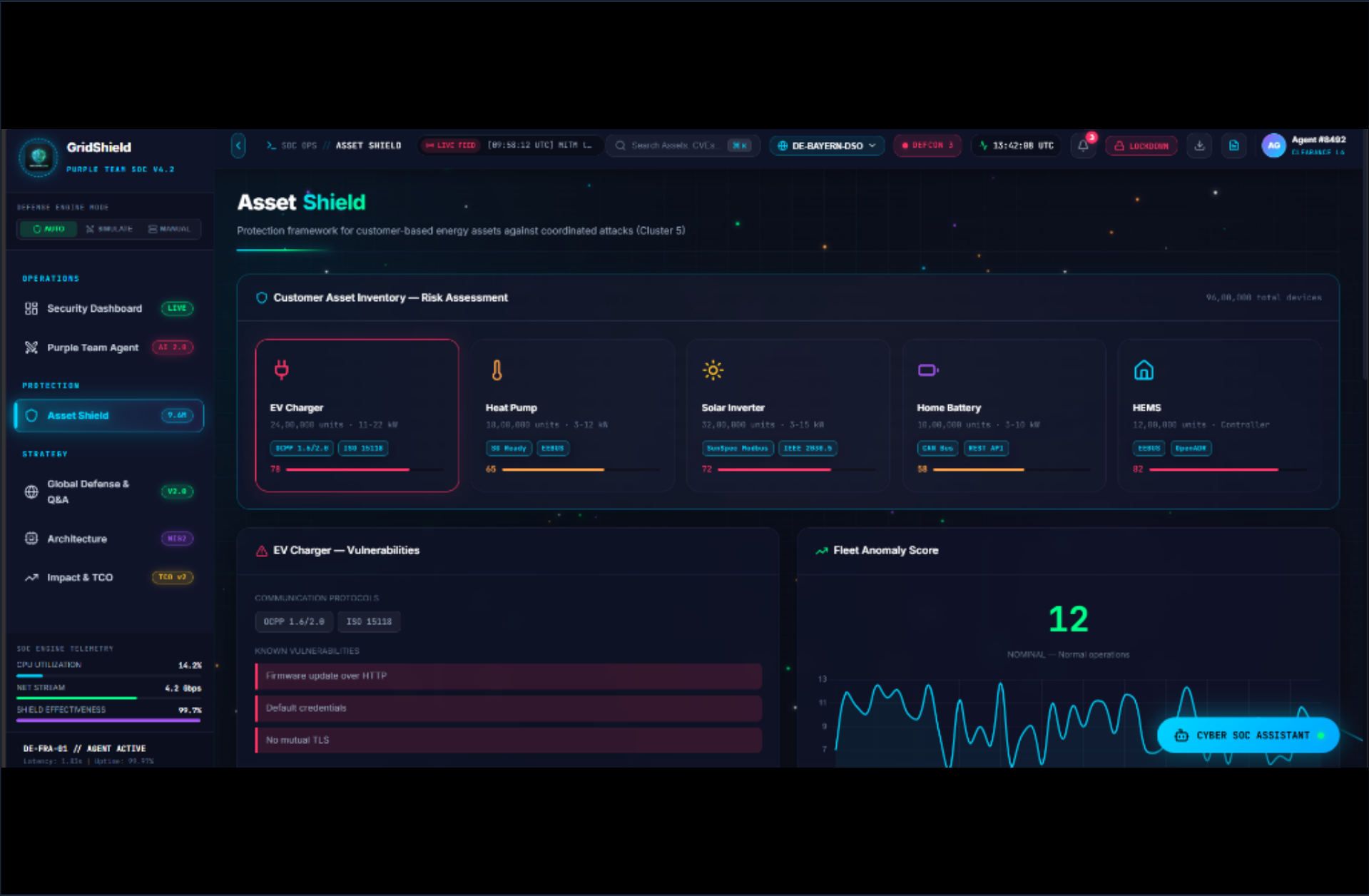
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Module 9: Customer DER Asset Shield — Inventory & Risk Matrix

Cluster 5: Protection Schemes for Customer-Based Assets

Decentralized protection matrix across 9.6M customer assets (EV Chargers, Heat Pumps, Solar Inverters, Batteries, HEMS).

Route: /asset-shield (Asset Registry)

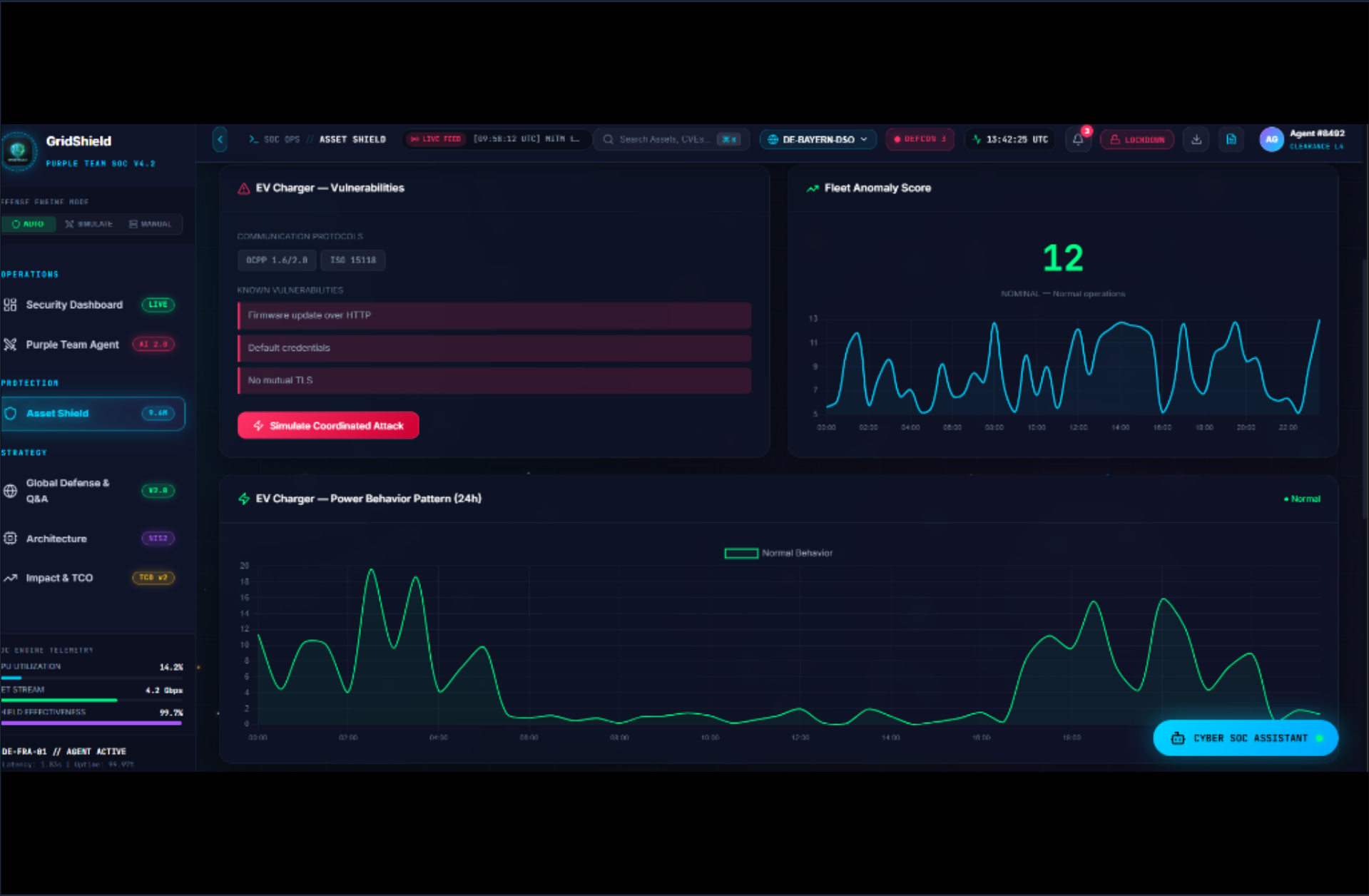


Module 10: Customer DER Asset Shield — Real-Time Telemetry & Anomaly

Live 24h EV charger power behavior curve and fleet anomaly score, validating on-device zero-trust execution under load surges.

Cluster 5: 1.14ms TinyML Anomaly Detection

Route: /asset-shield (Waveform Telemetry)

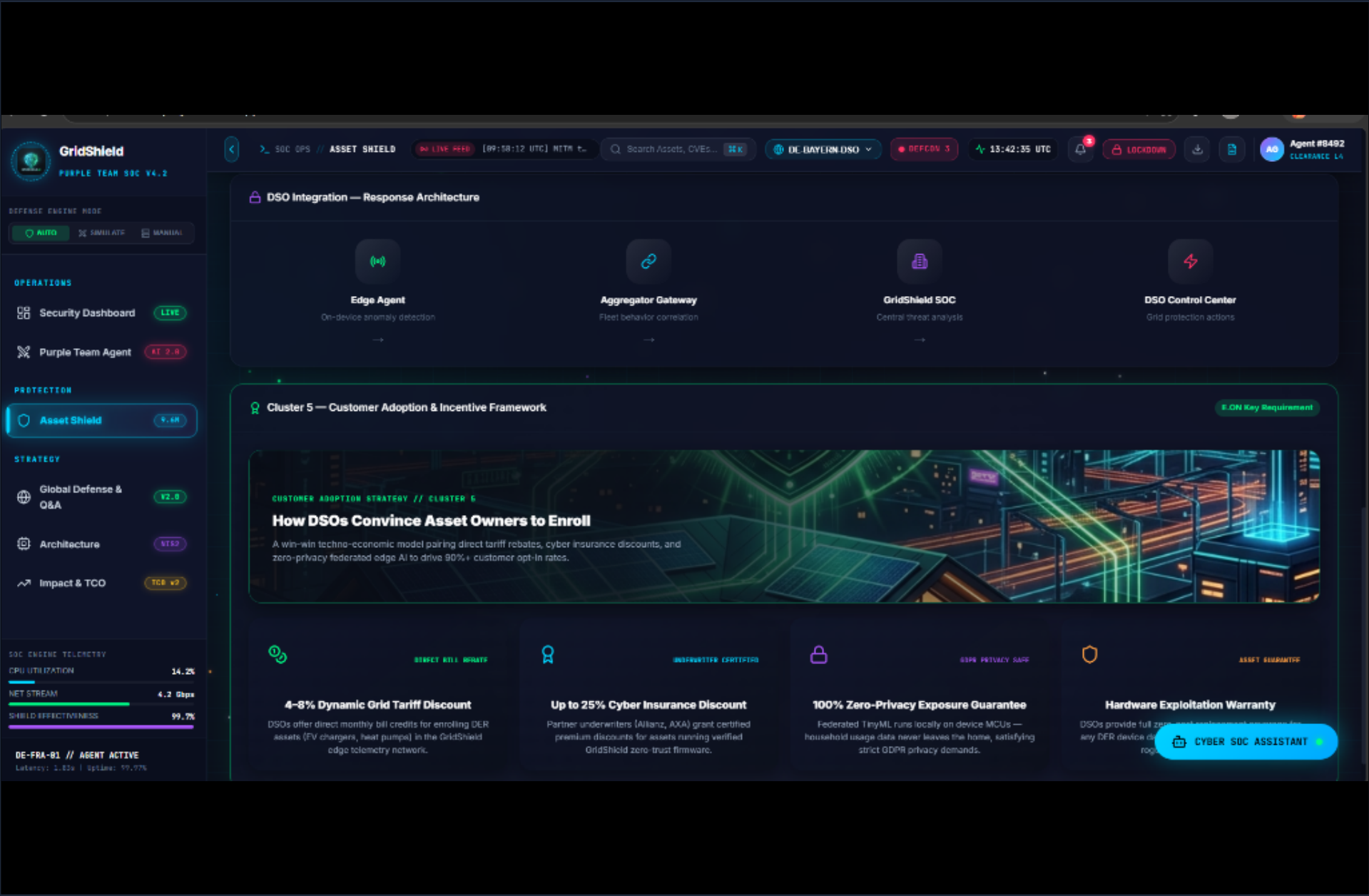


Module 11: Customer DER Asset Shield — 4-Pillar Incentive Framework

Cluster 5: Customer Adoption & Incentives

4-pillar incentive model: 4–8% dynamic tariff discount, up to 25% cyber insurance discount, 100% GDPR zero-leak guarantee, 100% GDPR zero-leak guarantee.

Route: /asset-shield (Incentive Model)



Global Scalability // US, Australia, ASEAN, EU

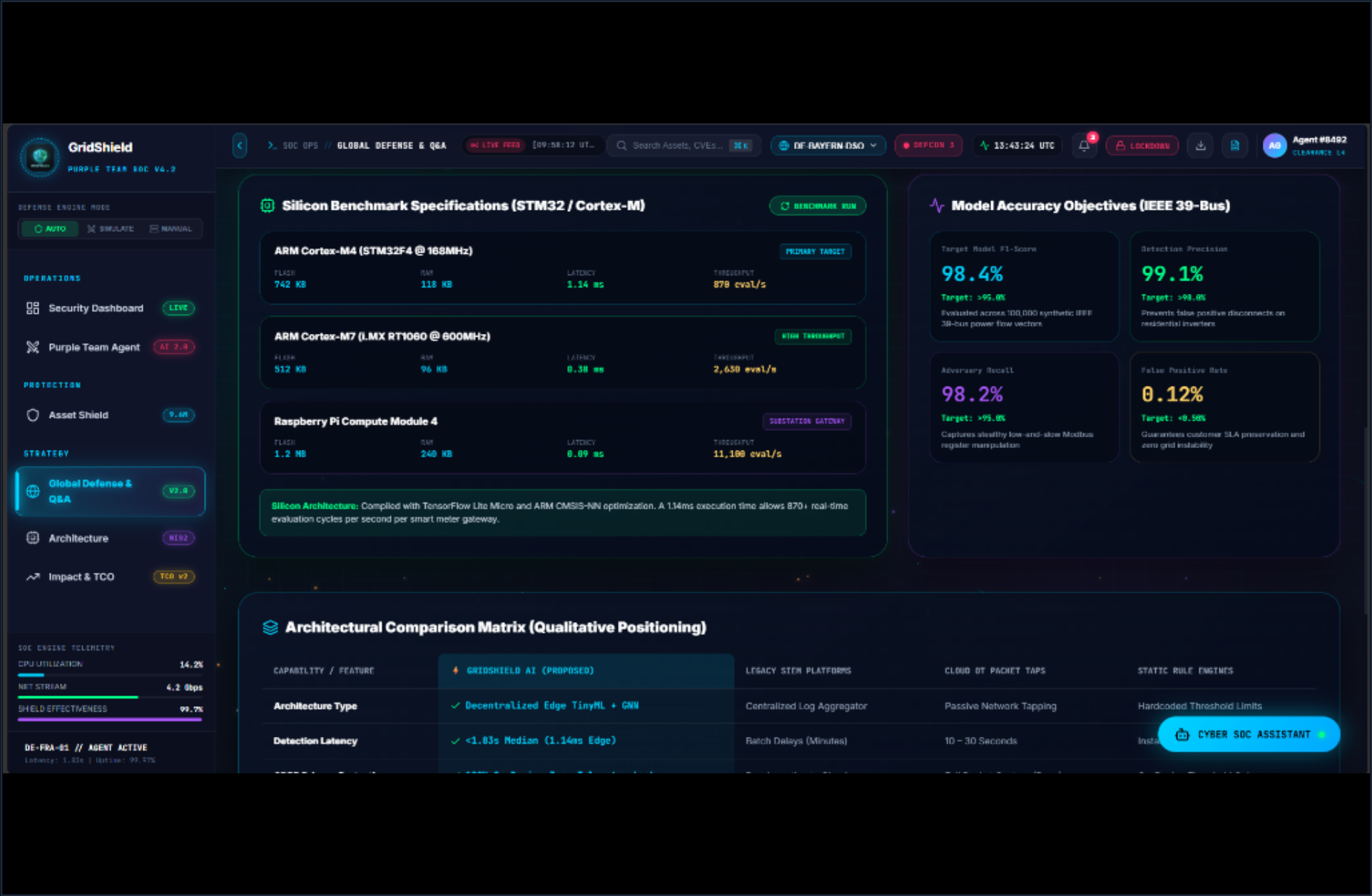
Route: /global-defense (Market Grid)

Module 13: Silicon Benchmarks & IEEE 39-Bus Accuracy Objectives

Silicon benchmarks on ARM Cortex-M4 (1.14ms, 118KB RAM) and Cortex-M7 with IEEE 39-bus targets (98.4% F1, 99.1% Precision).

Hardware Specifications & Model Performance

Route: /global-defense (Silicon Specs)



Module 14: Architectural Comparison & Technical Verification Protocols

Qualitative comparison against legacy SIEMs and mathematical proofs of GraphSAGE 2-hop sampling in <140ms on NVIDIA T4.

Judge Defensibility & Technical Verification

Route: /global-defense (Defensibility Proofs)



GridShield

PURPLE TEAM SOC V4.2

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9.9M

STRATEGY

Global Defense & Q&A

V2.0

Architecture

NIS2

Impact & TCO

TCO v2

EC ENGINE TELEMETRY

CPU UTILIZATION

14.3%

NET STREAM

4.2 Gbps

FIELD EFFECTIVENESS

99.7%

DE-FRA-01 // AGENT ACTIVE

SOC OPS // GLOBAL DEFENSE & Q&A

LIVE FEED (09:08:12 UTC)

Search Assets, CVEs...

DE-BAYERN-DSO

DEFCON 3

13:43:48 UTC

LOCKDOWN

Agent #B492

CLEARANCE L4

Architectural Comparison Matrix (Qualitative Positioning)

CAPABILITY / FEATURE	GRIDSHIELD AI (PROPOSED)	LEGACY SIEM PLATFORMS	CLOUD OT PACKET TAPS	STATIC RULE ENGINES
Architecture Type	✓ Decentralized Edge TinyML + GNN	Centralized Log Aggregator	Passive Network Tapping	Hardcoded Threshold Limits
Detection Latency	✓ <1.83s Median (1.14ms Edge)	Batch Delays (Minutes)	10 - 30 Seconds	Instant (High False Alarms)
GDPR Privacy Protection	✓ 100% On-Device Zero Telemetry Leak	Raw Ingestion to Cloud	Full Packet Capture (Pcap)	On-Device Threshold Only
Blast Radius Prediction	✓ PyTorch GraphSAGE 2-Hop Subgraphs	Manual SQL/SQL Queries	Basic IP/Subnet Groups	None (Isolated Endpoint)
TCO & Licensing Model	✓ Target €0.80 / device / year	Data Volume (€/GB Ingestion)	Hardware Appliance License	Negligible (Low Defense)

Security Architecture & Technical Verification Protocols

How does GridShield AI achieve sub-2-second GNN detection across 9.6M nodes without network congestion?

[VERIFIED PROOF]: GridShield AI does NOT stream raw 50Hz waveform telemetry from 9.6M devices to the cloud. 99.2% of raw telemetry is evaluated locally on device Cortex-M4 MCUs within 134ms. Only high-dimensional anomalous embeddings (risk score > 35) trigger upstream transmission. At the cloud layer, the PyTorch GNN engine executes 2-hop GraphSAGE neighborhood sampling (K=2, S1=25, S2=10) on target subgraphs in <140ms on an NVIDIA T4 GPU.

What are the itemized techno-economic assumptions behind the €0.80/device/year TCO baseline?

How does GridShield AI transfer to non-EU grids (US NERC-CIP, Australian NEM, ASEAN microgrids)?

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Module 15: Solution Architecture — Defense in Depth (Layers 1 & 2)

Enterprise OT/IT Zero-Trust Architecture

Layer 1 Edge Silicon (<1.14ms, <800KB) and Layer 2 Gateway Layer (10K msgs/sec, Modbus/OCPP/IEC 61850 protocol normalizer).

Route: /architecture (Layers 1 & 2)



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9.6M

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Global Defense & Q&A

V2.0

Architecture

R352

Impact & TCO

TCO V2

SOC ENGINE TELEMETRY

CPU UTILIZATION

14.2%

NET STREAM

4.2 Gbps

SHIELD EFFECTIVENESS

99.7%

DE-FRA-01 // AGENT ACTIVE

Latency: 1.14ms | Specimen: 99.9%

SOCC OPS // ARCHITECTURE

LIVE FEED (09:58:12 UTC) HITM t...

Search Assets, CVEs

DE-BAYERN-DSO

DEFCON 3

13:43:53 UTC

3

LOCKDOWN

AD Agent #2492 CLEARANCE L4

ENTERPRISE OT/IT ZERO-TRUST STACK // SPECIFICATION V2.0

Solution Architecture & Defense in Depth

4-Layer Zero-Trust Energy Defense combining 1.14ms Edge Silicon TinyML, Substation Gateway Normalizers, GraphSAGE GNN Blast Radius Prediction, and Automated SOAR Governance.

DEFENSE LAYERS

4 ACTIVE

SOAR AUTOMATION

92% PASS-THRU

FILTER DEFENSE LAYER:

ALL 4 LAYERS

LAYER 1: EDGE SILICON

LAYER 2: GATEWAY AGGREGATOR

LAYER 3: CLOUD GNN

LAYER 4: SOC SOAR

Layered Defense in Depth Architecture

ZERO-TRUST STACK

Layer 1: Edge Silicon — On-Device TinyML

1.14ms Microcontroller Defense

Latency: <1.14ms | Binary: <800KB | RAM: 118KB | Power: 1.8mW

Lightweight C++ AI agents compiled with CMSIS-NN running directly on customer energy assets and smart meter gateways. Performs real-time waveform telemetry classification and zero-trust firmware attestation.

INTEGRATED MODULES & HARDWARE AGENTS:

TinyML Inference Engine

Firmware Attestation

Local Anomaly Detector

Secure Boot Chain

OTA Delta Manager

Layer 2: Gateway Layer — Multi-Protocol Aggregation

10K msgs/sec Fleet Correlation

Throughput: 10K msgs/sec | Protocols: 6 Standards | Buffer: 72 Hours

Substation and aggregator gateways correlating behavior across distributed device fleets. Normalizes heterogeneous industrial protocols (Modbus TCP, OCPP 2.0.1, IEC 61850) with store-and-forward.

INTEGRATED MODULES & HARDWARE AGENTS:

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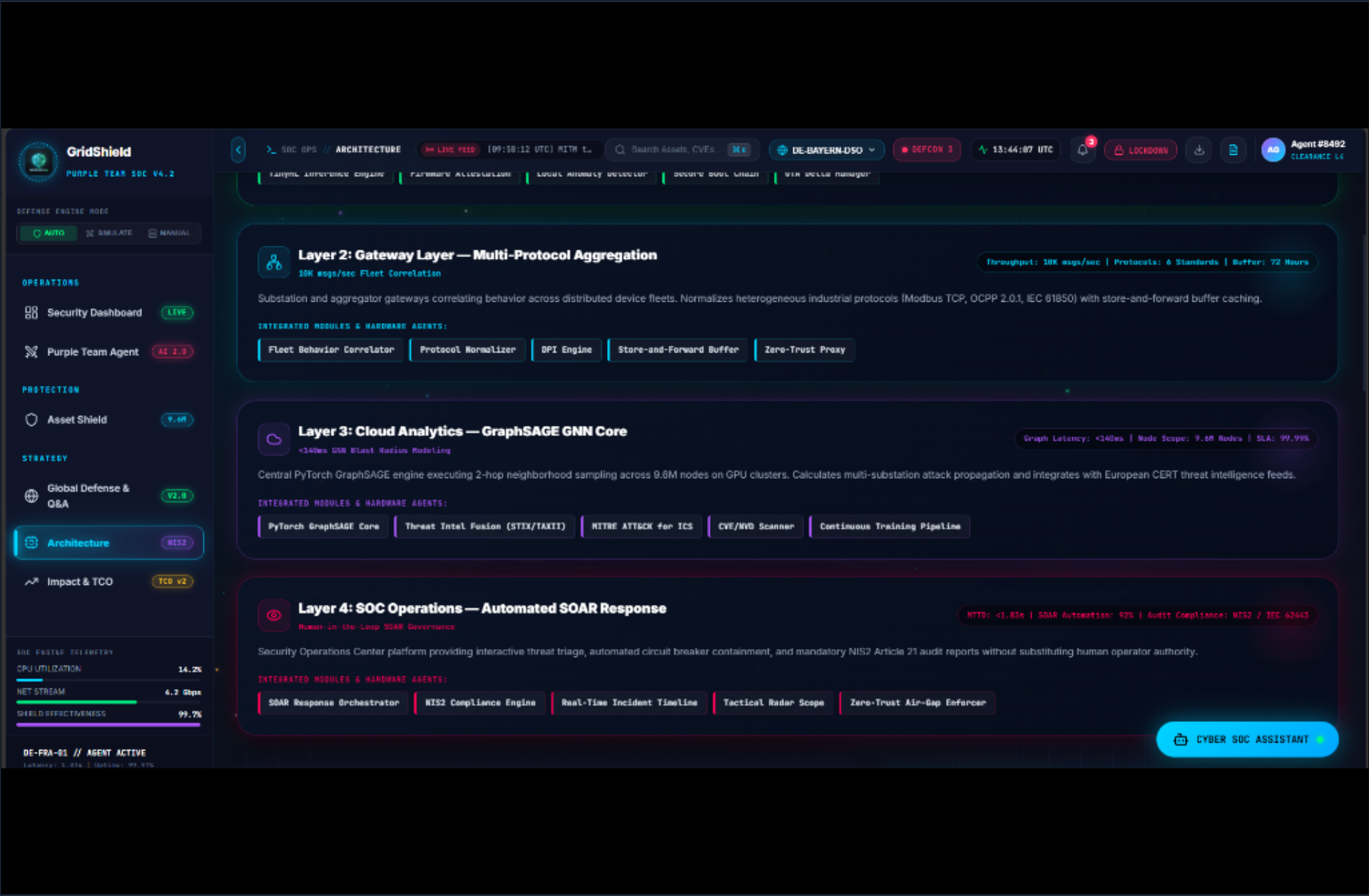
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Module 16: Solution Architecture — Cloud GNN & SOC SOAR (Layers 3 & 4)

Enterprise OT/IT Zero-Trust Architecture

Layer 3 Cloud Analytics (PyTorch GraphSAGE 2-hop GNN, STIX/TAXII threat intel) and Layer 4 SOC SOAR (<1.83s MTTD, 92% pass-thru).

Route: /architecture (Layers 3 & 4)

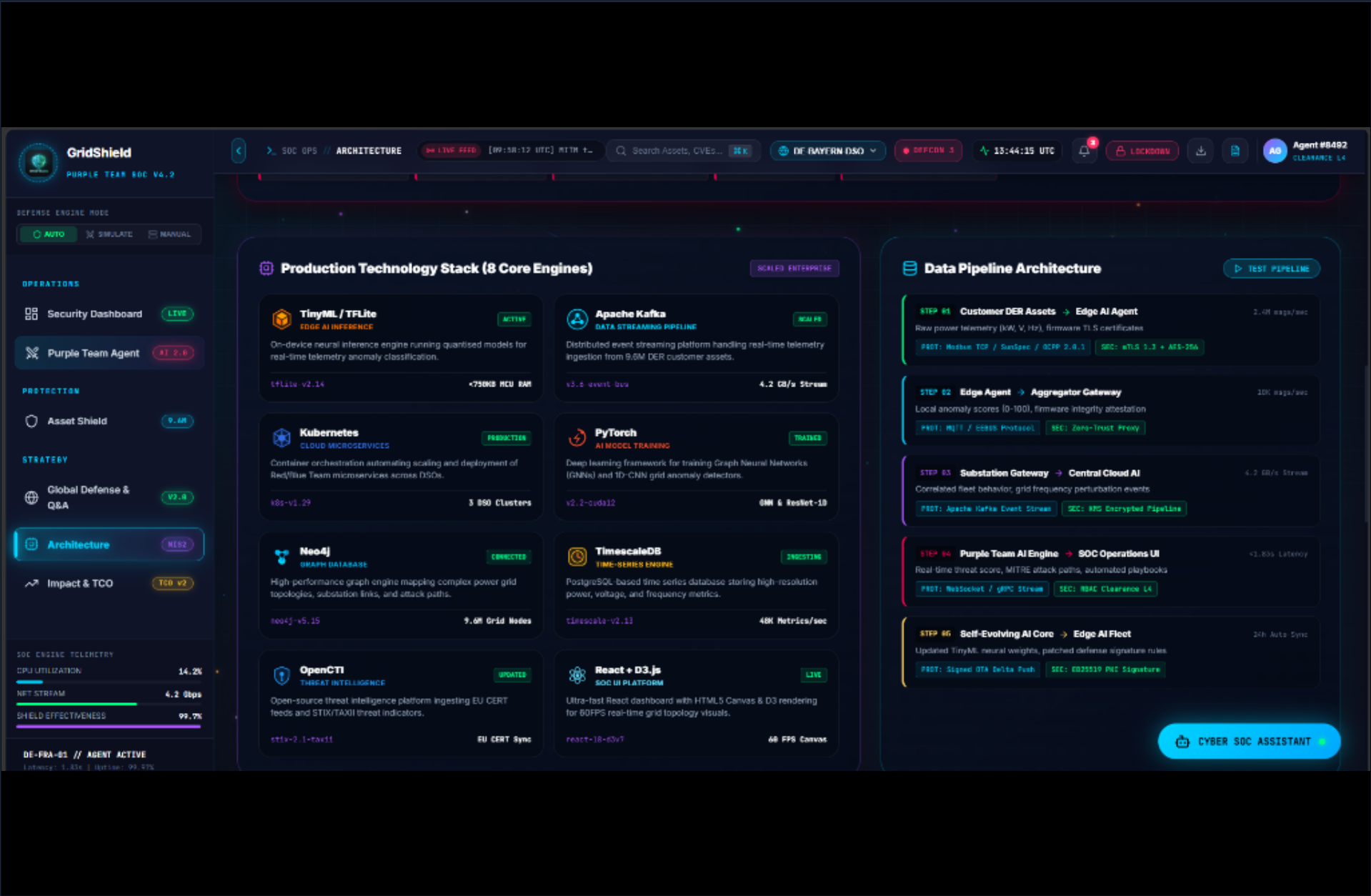


Module 17: Production Tech Stack & 5-Step Data Pipeline Architecture

Production stack (TinyML, Kafka, Kubernetes, PyTorch GNN, Neo4j, TimescaleDB) and interactive data pipeline tester.

Production Engineering // 8 Core Engines

Route: /architecture (Tech Stack & Pipeline)

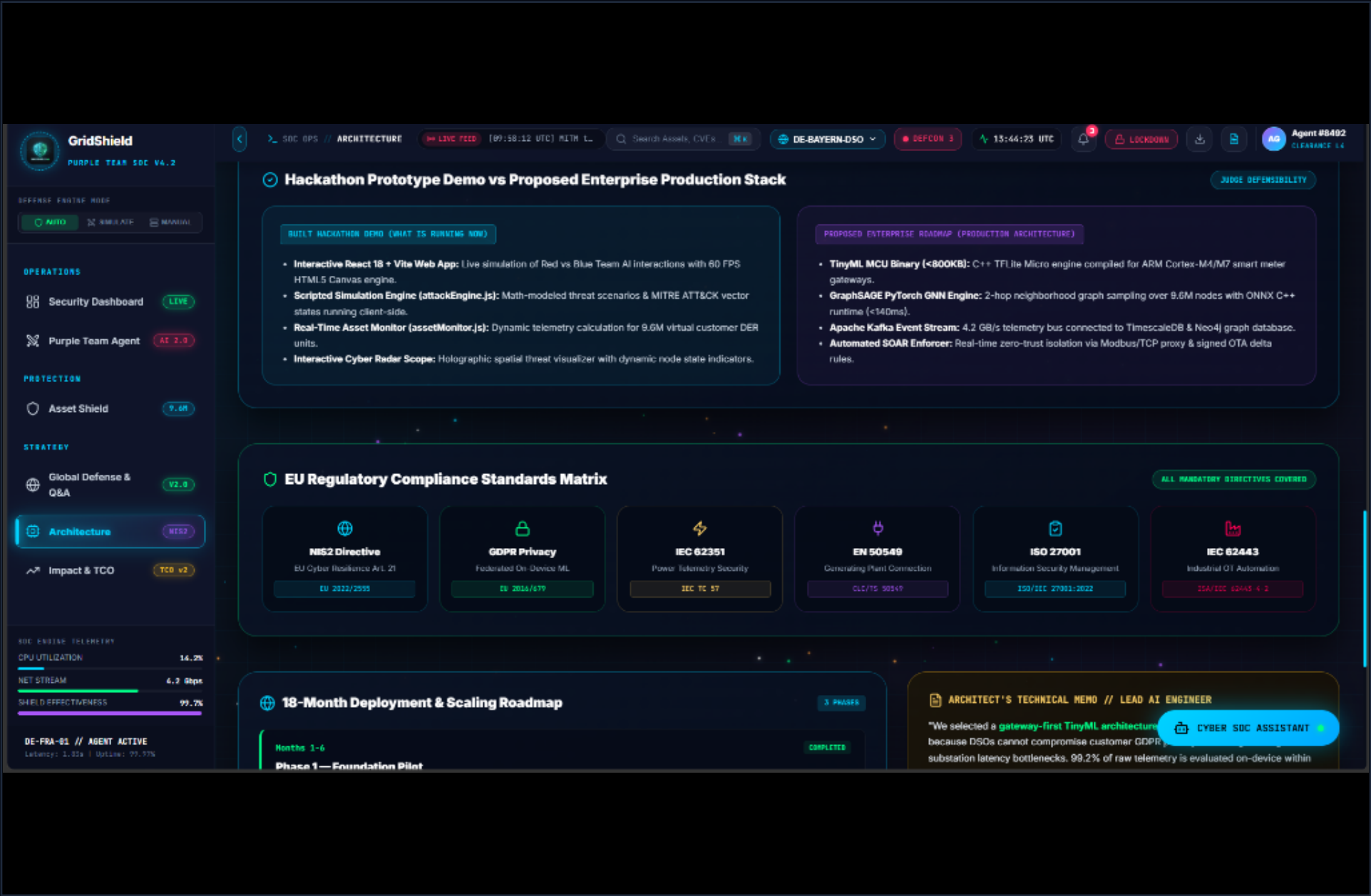


Module 18: Prototype vs Enterprise Stack & EU Regulatory Matrix

Prototype vs enterprise roadmap comparison and full coverage matrix for EU NIS2, GDPR, IEC 62351, EN 50549, and ISO 27001.

Compliance // NIS2, GDPR, IEC 62351, IEC 62443

Route: /architecture (Compliance Matrix)



Program Rollout & Executive Architecture Memo

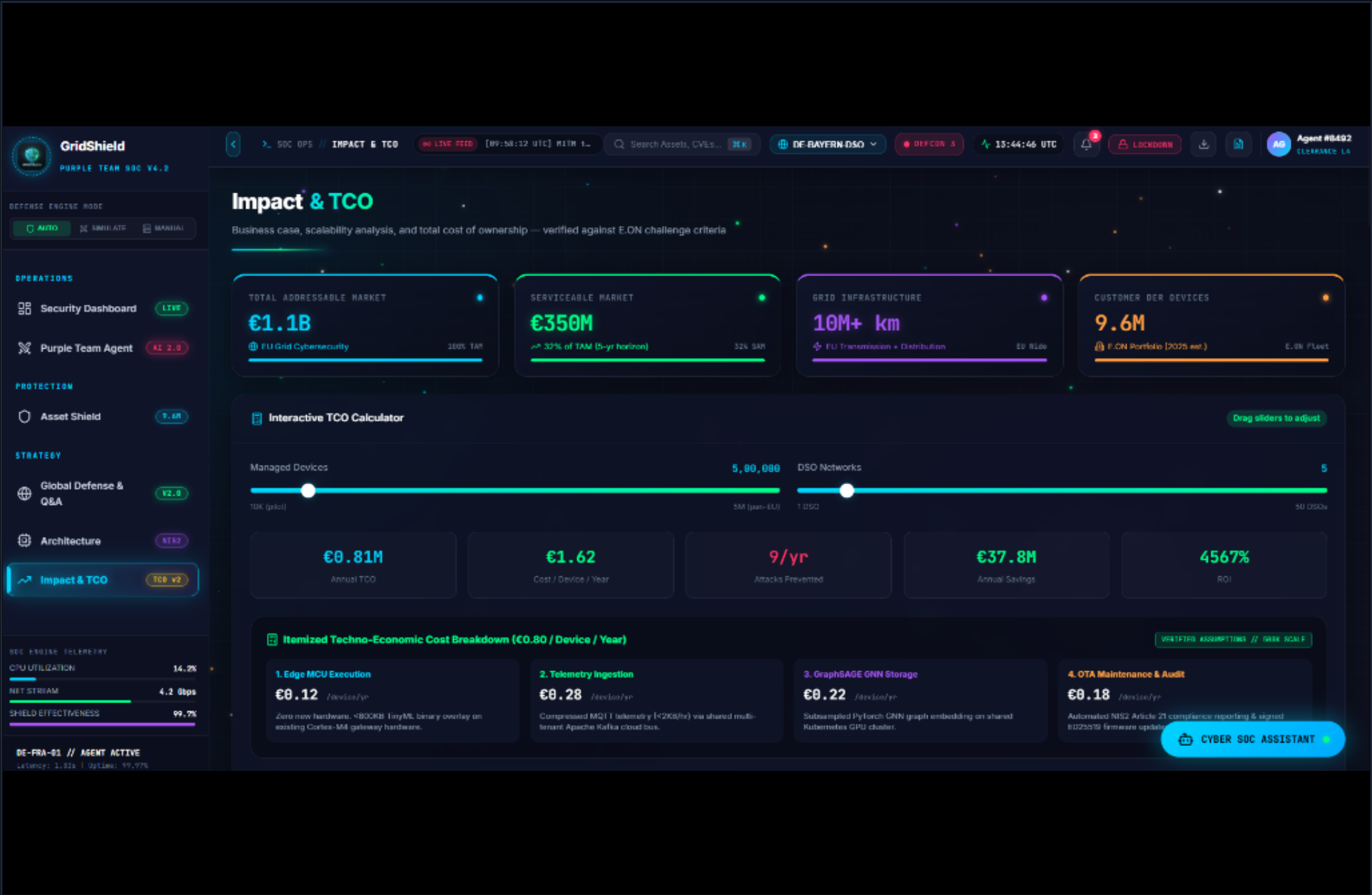
Route: /architecture (Roadmap & Memo)

Module 20: Impact & TCO — Market Opportunity & Interactive TCO Calculator

Frugal Engineering // €0.80/device/year Baseline

Executive market sizing (€1.1B TAM, €350M SAM, 9.6M DER fleet) and dynamic TCO slider modeling €37.8M annual savings.

Route: /impact (TCO Calculator)

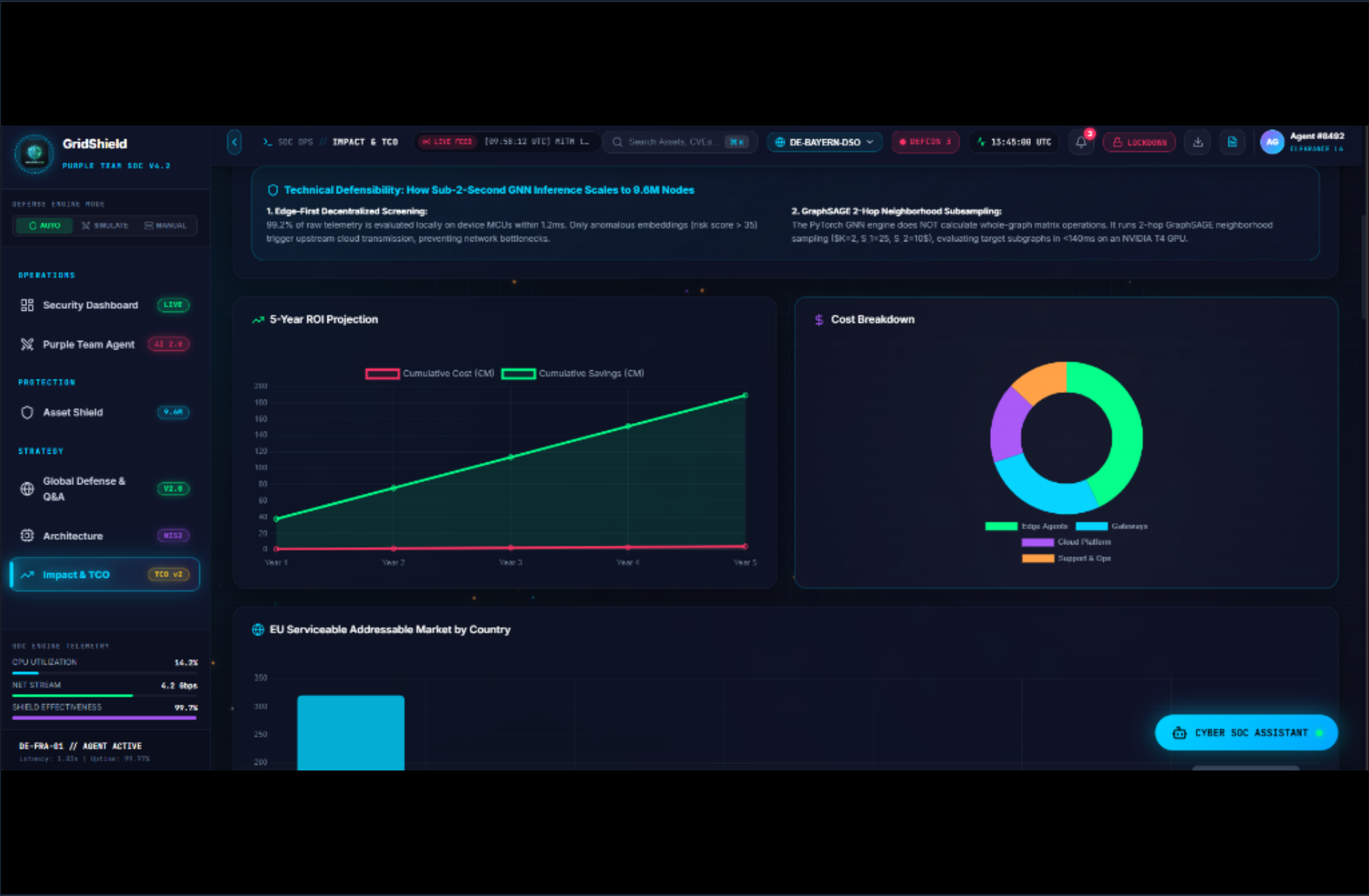


Module 21: Impact & TCO — GNN Defensibility & 5-Year ROI Projection

Financial Validation & Sub-2s GNN Scaling

Mathematical proof of sub-2s GNN scaling across 9.6M nodes, 5-year cumulative ROI projection curve, and cost breakdown donut.

Route: /impact (ROI Projections)

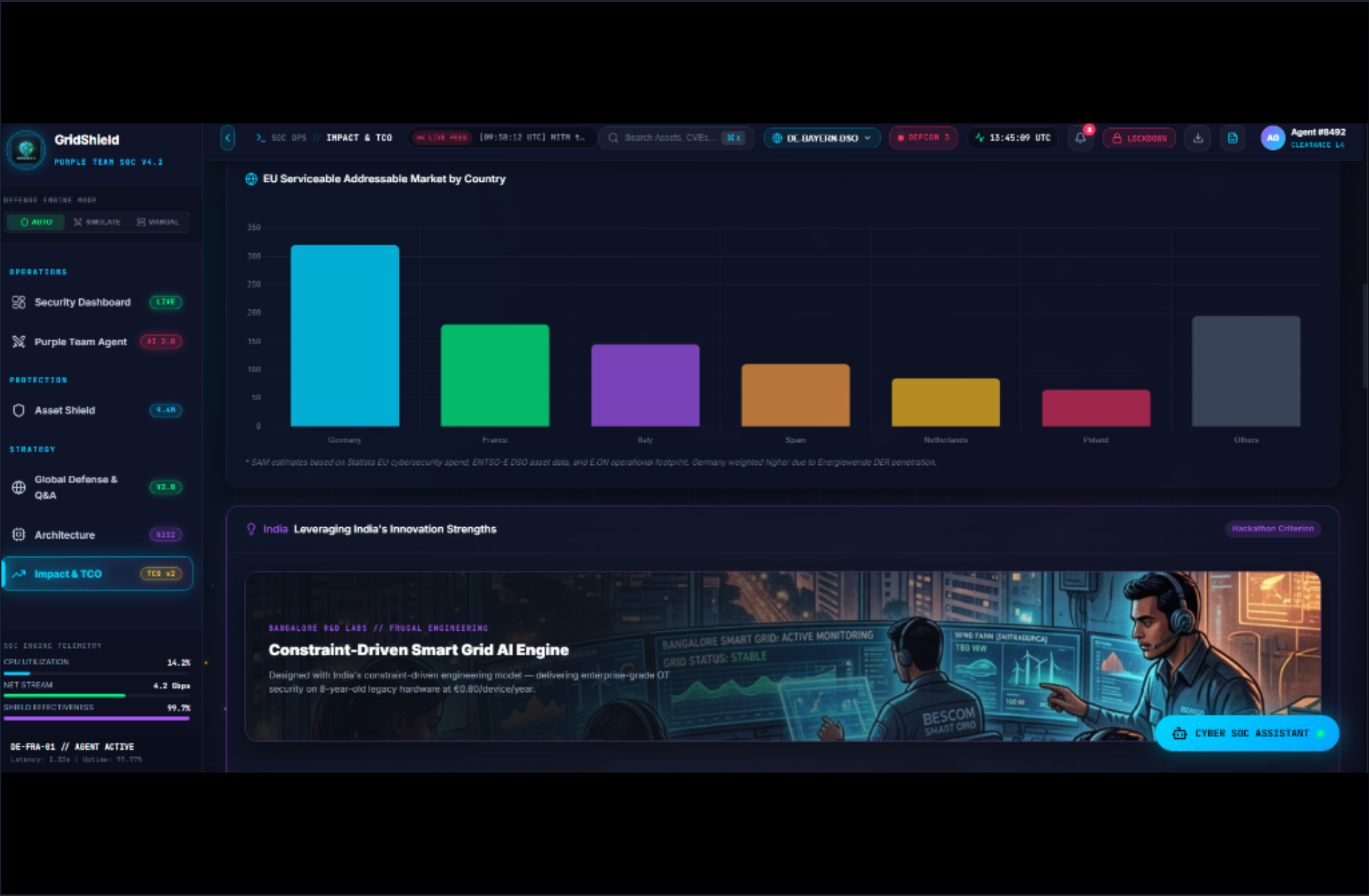


Module 22: Impact & TCO — EU Country SAM & Indian Innovation Strengths

European country-by-country addressable market and Bangalore R&D Labs spotlight on constraint-driven engineering.

Criteria for All Challenges // Indian Innovation

Route: /impact (Global Expansion & India Strengths)



Module 23: Impact & TCO — Frugal Mindset & Quantified Impact Metrics

Frugal innovation pillars, Lead Architect verified quote badge, and quantified impact metrics: 450M+ protected households.

Quantified Impact // 450M+ EU Households

Route: /impact (Frugal Engineering & Impact)

